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Honours Individual Project Dissertation

MUSIC COMPASS: DESIGNING AND EVALUATING A SPATIAL-AUDIO INTERFACE FOR NON-VISUAL MUSIC BROWSING ON MOBILE DEVICES

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Abstract

The ubiquitous reliance on the smartphone's visual "Glass Interface" for navigating digital music libraries introduces severe situational impairments in nomadic environments and creates a problematic accessibility bottleneck while multitasking on the move and for impaired visual users. To address this, this dissertation proposes and evaluates the *Music Compass*: a purely auditory, eyes-free interface that leverages human proprioception and spatial memory.

A Progressive Web App (PWA) was developed utilizing the HTML5 DeviceOrientation and Web Audio APIs to map a hierarchical music library (Genres, Artists, Tracks) to a continuous 360° spatial soundscape anchored to the user's physical body. To test the theoretical boundaries of gestural control, an $N = 12$ within-subjects evaluation was conducted comparing two interaction paradigms: a continuous "Always-On" tracking mode, and a "Clutch" (hold-to-browse) mode.

Results indicated that while the Always-On mode yielded statistically faster target acquisition ($p < 0.001$), participants overwhelmingly rejected it in favor of the Clutch mode. This revealed a clear example of the "Ergonomics of Agency," demonstrating that users prioritize explicit, tactile pause states over raw efficiency to avoid the "Midas Touch" problem. Furthermore, the evaluation proved the rapid acquisition of spatial motor learning, with retrieval times dropping by over 50% across three trials. Finally, observational data highlighted critical design constraints for future spatial audio systems, including the acoustic masking of commercial track intros and the impact of difficult transit environments.

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Euan, for his invaluable guidance, continuous feedback, and for pushing me to critically examine the HCI theories beneath the code throughout this academic year.

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Finally, a special thank you to my dad for his continuous support and encouragement.

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Signature: Calum Woods Date: 20 March 2026

Contents

1	Introduction	1
1.1	Evolution of Music Consumption	1
1.2	Motivation	1
1.3	Research Goals	2
1.3.1	Research Questions	2
1.4	Summary of Contributions	2
1.5	Scope and Limitations	3
2	Background & Literature Review	4
2.1	The Visual Bottleneck in Nomadic HCI	4
2.2	The Psychology of Browsing: Search vs. Discovery	4
2.3	Spatial Audio and Cognitive Limits	4
2.3.1	Auditory Masking and Distinctiveness	5
2.3.2	Spatial Resolution and Memory	5
2.3.3	Precedents in Spatial Audio and Non-Visual Browsing	5
2.4	Interaction Modalities: Theoretical Foundations	5
2.4.1	Proprioception and Spatial Motor Learning	6
2.4.2	The Midas Touch Problem and Buxton's Three-State Model	6
2.4.3	Gross Motor Gestures and SATO	6
3	Design & System Development	7
3.1	System Overview and Interaction Design	7
3.1.1	Hierarchical Navigation and the "Vertical Slice" Strategy	8
3.1.2	Visual Interface and Experimental Scaffolding	8
3.2	The Two Navigational Modalities	9
3.2.1	Condition A: The "Clutch" Mode (Hold-to-Browse)	10
3.2.2	Condition B: The "Always-On" Mode (Continuous Tracking)	10
3.3	Auditory Design Rationale & Iterative Redesigns	11
3.4	Technical Implementation	11
3.4.1	Dynamic Calibration and Stereo Panning	12
3.4.2	Sensor Integration and Gesture Detection	12
4	Evaluation Methodology & Results	14
4.1	Evaluation Methodology	14
4.1.1	Objectives	14
4.1.2	Participants	14
4.1.3	Experimental Design	14

4.1.4	Tasks & Apparatus	14
4.1.5	Pre-Task Scaffolding and The Soundboard	14
4.1.6	The Visual “Stop” Button	15
4.1.7	Metrics Collected	15
4.2	Quantitative Results	16
4.2.1	Efficiency and Task Completion Times	16
4.2.2	The Learning Effect and Spatial Mental Mapping	17
4.2.3	Error Rates	18
4.3	Qualitative Results	19
4.3.1	Physical Fatigue and Interaction Ergonomics	19
4.3.2	Perceived Controllability and the Ignored Stop Button	19
4.3.3	The Efficacy of the Soundboard	20
4.3.4	Gamification and Entertainment Value	20
4.3.5	Feedback on Gesture Mechanics	20
5	Discussion & Future Work	21
5.1	Synthesizing Usability & Performance	21
5.1.1	Gross Motor Gestures and the Speed-Accuracy Trade-Off (SATO)	21
5.1.2	Proprioception and Spatial Motor Learning	21
5.2	Affordances, Agency, and Mental Models	21
5.3	Context of Use & Unanticipated Outcomes	22
5.3.1	Auditory Masking vs. Musical Familiarity	22
5.3.2	Nomadic HCI and Environmental Interference	22
5.3.3	Situational Awareness and Physical Adaptations	23
5.4	Future Work	23
6	Conclusion	25
	Appendices	27
A	Evaluation Materials	27
B	Prototype Interface Gallery	33
C	System Deployment & Readme	36
C.1	Source Code Overview	36
C.2	How to Run the Prototype	36
	Bibliography	37

1 | Introduction

1.1 Evolution of Music Consumption

The relationship between the listener and their music library has undergone a radical transformation over the last three decades. Over that time, there was an accelerating transition away from the tactile physical media era, such as CDs, vinyl records, and cassette tapes. Music library curation, selection, and playback involved those individual physical recorded objects, typically stored together like books in a library, picked by sight and played individually by physical loading onto a dedicated player. We have now transitioned to an era of ubiquitous cloud streaming. Platforms such as Spotify and Apple Music provide instant access to practically infinite repositories of content, accessed via a wide variety of electronic playback devices and curated virtually with playlists and "liked" links, or searched for in real time. While this shift has democratized access, massively increasing the depth of choice and access to recorded music, it has simultaneously introduced a usability paradox (Schwartz 2004).

The primary paradigm for interacting with these vast digital repositories remains the high-resolution visual display—a “glass computer screen” with a graphical user interface (GUI) that demands focused foveal attention and precise fine-motor control. While this is highly effective to manage musical playback while static, sitting at a desktop or standing still on a phone, this visual hegemony reveals a critical bottleneck in eyes-free mobile scenarios. In contexts where the user’s visual attention is otherwise engaged by the physical environment (e.g., navigating a busy urban street, driving, or jogging), the requirement to look at a screen is not merely an inconvenience; it represents a severe situational impairment and potentially a physical safety hazard (Sarsenbayeva et al. 2017).

Furthermore, this reliance on visually navigated control mechanisms systematically disenfranchises blind and low-vision users. Current accessibility solutions, primarily screen readers like VoiceOver, attempt to address the problem by linearizing the visual interface into a sequential auditory stream. As explored in later chapters, this approach forces users into a linear, sequential search pattern. When navigating large hierarchical datasets, this serial traversal imposes a massive cognitive load on short-term memory, turning the joy of searching for an individual well-loved track or spontaneous music discovery into a tedious, high-effort exercise. Consequently, users in non-visual contexts are often forced to abdicate control, surrendering their agency to opaque recommendation algorithms or passive “shuffle” modes (Kamalzadeh et al. 2016).

1.2 Motivation

This research proposes that the solution to the eyes-free browsing problem lies not in refining visual lists or improving standard voice recognition, but in reimagining the fundamental interaction paradigm entirely. The human capacity for spatial hearing (localizing sound in 3D space) and proprioception (the innate sense of body orientation) offers a high-bandwidth, intuitive channel for navigation that remains heavily underutilized in mobile Human-Computer Interaction (HCI) (Lopes et al. 2015).

The vision for this project, the *Music Compass*, is to map the abstract hierarchy of a music library (Genres, Artists, Tracks) to the physical space surrounding the user’s body, creating a hierarchical spatial audio menu. In this proposed organisational framework, “Rock” might reside to the North and “Jazz” to the East. Navigation becomes an act of physical orientation—turning the body to “tune in” to a genre—rather than an act of visual scanning.

However, transitioning from a rigid visual interface to a continuous gestural one introduces significant theoretical challenges regarding user agency. The central tension this project aims to investigate is the balance between immersion and precision. If the system is “Always-On,” responding to every physical movement, it risks inducing the “Midas Touch” problem, where natural environmental movements are misinterpreted as explicit commands (Jacob 1990). Conversely, if the system requires strict “Clutching” (e.g., holding a physical or digital button to explicitly allow movement tracking), it may introduce localized physical fatigue and possible limitations for motor-impaired individuals.

1.3 Research Goals

The overarching goal of this dissertation is to design, implement, and formally evaluate a non-visual prototype of a hierarchical spatial audio menu. This system should leverage modern smartphone Inertial Measurement Units (IMUs) and the Web Audio API to test these spatial interaction theories in a functional environment. Rather than focusing solely on the acoustic playback of audio, this research focuses deeply on the physical control mechanisms required to make such navigation possible and usable in real-world scenarios where sighted GUI access is difficult or dangerous.

The project evaluates two distinct interaction modalities to determine which offers the optimal balance of efficiency and perceived user control:

1. **Explicit Control (The Clutch):** A modality where the user must actively hold the screen to engage the gyroscope compass, providing a definitive pause state when released.
2. **Continuous Tracking (Always-On):** A modality where the compass is permanently active and reading sensor data, requiring only a discrete “tap” action to select an item.

1.3.1 Research Questions

To empirically guide the design and evaluation of the *Music Compass*, the following Research Questions (RQs) were established:

- **RQ1 (Efficiency and Target Acquisition):** Which interaction paradigm, Explicit Control (Clutch) or Continuous Tracking (Always-On), results in statistically faster target acquisition within a hierarchical spatial audio menu?
- **RQ2 (Usability and Agency):** How does the presence or absence of an explicit “pause” state (the Clutch) alter a user’s perceived sense of control and their physiological fatigue (e.g., localized grip tension versus postural arm strain) during prolonged spatial browsing?
- **RQ3 (Error Recovery and SATO):** How does the inclusion of a gross-motor error-recovery mechanic (a physical device “Shake”) impact the user’s navigational pacing and their adherence to the Speed-Accuracy Trade-off (SATO)?

1.4 Summary of Contributions

Hands-free, screen-free interaction is increasingly desirable. This paradigm is essential not just for visually impaired users, but as a lifestyle choice for multitasking pedestrians who wish to

remain engaged with their physical surroundings. The ubiquity of wireless earbuds and smart speakers means users frequently consume audio with their device out of sight or out of reach.

However, current Voice User Interfaces (VUIs) remain largely “one-shot” query systems (e.g., “Play Nirvana”). These systems fail to support exploratory browsing because they demand specific *recall* memory from the user, rather than providing options for *recognition*. In parallel, HCI research indicates that users increasingly desire greater agency and self-customization, wishing to escape the “black-box” nature of algorithmic playlists in favor of transparent, user-steered navigation (Kamalzadeh et al. 2016).

This dissertation contributes to the growing field of eyes-free mobile HCI by providing a functional, browser-based proof-of-concept for spatial audio navigation. Furthermore, through an $N = 12$ empirical user study, this report contributes critical insights into the limitations of continuous sensor tracking in transit environments. In particular, it highlights the requirement for distinct acoustic track transitions and proves the ergonomic necessity of explicit exit states in gesture-based UI.

1.5 Scope and Limitations

It is important to clearly define the boundaries of this research. The *Music Compass* focuses entirely on Human-Computer Interaction (HCI) and the physical control mechanisms of spatial audio. Consequently, the development of novel Music Information Retrieval (MIR) feature extraction, or AI-based recommendation algorithms, is strictly out of scope. Incorporating opaque machine learning models would actively compromise the project’s focus on user agency and system transparency. Therefore, the prototype assumes a pre-categorized library, isolating the evaluation purely to the individual spatial interaction modalities and the theoretical contrast between them.

2 | Background & Literature Review

To ground the research and identify the specific knowledge gap this project addresses, this chapter synthesizes three distinct threads of Human-Computer Interaction (HCI) literature: (1) The cognitive limitations of visual and sequential retrieval systems in mobile environments; (2) The psychoacoustic foundations of spatial audio and auditory masking; and (3) The theoretical mechanics of gestural interaction and proprioception.

2.1 The Visual Bottleneck in Nomadic HCI

The evolution of human-computer interaction has been heavily impacted by the transition from static desktop computing to dynamic, unpredictable mobile environments. Traditional graphical user interfaces (GUIs) were engineered under the assumption that the user could dedicate their full visual and cognitive attention to a screen. However, ubiquitous mobile computing has fractured this assumption. When individuals interact with mobile devices while simultaneously navigating physical environments (e.g., walking to a bus), they encounter a severe sensory bottleneck (Sarsenbayeva et al. 2017).

Navigating the physical world is inherently a visual-spatial and motor-intensive task. Forcing a user to divide their finite visual attention between a digital screen and the physical environment introduces a dangerous cognitive overload. The demand for “eyes-free” interaction arises directly from the need to preserve pedestrian situational awareness and physical safety (Kane et al. 2009).

2.2 The Psychology of Browsing: Search vs. Discovery

It is critical to distinguish between *searching* (retrieving a specific, known item) and *browsing* (exploring an information space to find something satisfying, with no predefined target). Voice User Interfaces (VUIs) like Siri or Alexa excel at the former because they are query-driven systems that rely entirely on the user’s *recall* memory. However, they fail systemically at the latter. Facing an immense database without a structured way to explore it leads to cognitive overload and “choice paralysis” (Schwartz 2004).

Current non-visual tools force users into a rigid “Search” mindset because they lack the bandwidth to support open-ended exploration. Screen readers, by linearizing lists, impose a heavy cognitive load on the user’s phonological loop (the component of working memory responsible for processing auditory information). Users cannot quickly “scan” audio as they do a visual list; they must wait for serial enunciation. These limitations can only be mitigated via a novel interface that provides a way to “glance” at audio data by mapping it spatially, allowing users to navigate multiple dimensions simultaneously rather than sequentially.

2.3 Spatial Audio and Cognitive Limits

To bypass the visual bottleneck, the *Music Compass* relies on the brain’s innate ability to map sound to physical space. However, the biological limits of human psychoacoustics heavily dictate

the feasible boundaries of the system architecture.

2.3.1 Auditory Masking and Distinctiveness

Early concepts in auditory displays often attempted to utilize the “Cocktail Party Effect”—the human ability to focus auditory attention on a single stimulus while filtering out a range of concurrent background noises. However, HCI literature warns that presenting multiple concurrent audio streams quickly exceeds the user’s processing capacity.

This leads to a phenomenon known as *auditory masking* (McGookin and Brewster 2004). Auditory masking occurs when concurrent audio streams share similar frequency bands, causing louder or more complex sounds to neurologically obscure softer ones. Without visual cues to anchor the sound sources, the user’s auditory cortex struggles to isolate individual tracks, resulting in severe spatial confusion.

Furthermore, utilizing raw commercial audio (such as the slow, instrumental build-up of a pop track) fails to provide immediate “auditory distinctiveness.” This lack of immediate recognition causes cognitive delays, extending the navigational process and ultimately leading to user fatigue or abandonment (McGookin and Brewster 2004). Therefore, effective spatial interfaces heavily favour a sequential panning approach, utilizing highly distinct, synthesized cues (or “Earcons”) to allow for rapid, recognizable scanning (Walker et al. 2006).

2.3.2 Spatial Resolution and Memory

A critical constraint for spatial design is the resolution of human spatial memory. Research by Brewster et al. (Brewster et al. 2003) suggests that for egocentric auditory menus, the upper biological limit of reliable spatial differentiation is approximately eight distinct sectors (roughly 45° each). This implies that a system cannot simply map a list of 100 artists to a single circle without causing severe overlap. The interface must be structured hierarchically, implementing a form of “Semantic Zooming,” where selecting a parent sector (e.g., a Genre) dynamically loads a new “ring” of children nodes (e.g., Artists) into the same 360-degree space.

2.3.3 Precedents in Spatial Audio and Non-Visual Browsing

Prior research has demonstrated the viability of non-visual audio navigation. Hiipakka et al. successfully utilized stereo panning to enable eyes-free navigation of a hierarchical playlist menu (Hiipakka and Lorho 2003). Similarly, Microsoft’s *Soundscape* project proved that visually impaired users could build rich, accurate mental maps of real-world environments using 3D spatial audio cues (Microsoft Research 2018).

In the domain of music exploration, Bergman’s *Hands-On Music* and Keränen’s *Gravity Sphere* utilized device tilt and orientation to browse songs mapped to a 3D space (Keränen et al. 2009). However, these early iterations often lacked user-centric metadata organization and forced a rigid, high-effort physical posture. Furthermore, while systems like Kamalzadeh’s *TagFlip* successfully proved that lightweight, user-steered navigation increases perceived control and satisfaction over algorithmic recommendations (Kamalzadeh et al. 2016), they remained bound to visual screens. The *Music Compass* synthesizes these insights, combining the bodily spatial mapping of *Gravity Sphere* with the transparent, high-agency design philosophy of *TagFlip*.

2.4 Interaction Modalities: Theoretical Foundations

The core contribution of this project is evaluating exactly *how* users physically control this spatial exploration and the psychological impact of those control mechanisms.

2.4.1 Proprioception and Spatial Motor Learning

A fundamental advantage of spatially anchored interfaces is their ability to leverage proprioception, the human body's subconscious awareness of its own posture and limb placement (Lopes et al. 2015). By mapping digital audio nodes to fixed physical angles around the user's body, the interface allows the user to build an egocentric spatial mental map. Over time, the cognitive reliance on listening for audio feedback is naturally replaced by physical muscle memory, significantly increasing interaction speed (Lopes et al. 2015).

2.4.2 The Midas Touch Problem and Buxton's Three-State Model

The "Midas Touch" problem, originally coined in eye-tracking research (Jacob 1990), refers to the difficulty of distinguishing between moving for *perception* (looking around the environment) and moving for *action* (executing a digital command). In an orientation-based music player, if the user turns their body to check for traffic, an "always-on" tracking system will likely misinterpret this natural environmental movement as a deliberate command to change the track.

To formalize this mechanical tension, this research applies Buxton's Three-State Model of graphical input (Buxton 1990):

- **State 0 (Out of Range):** The system is idle and ignoring input.
- **State 1 (Tracking):** The system tracks motion but does not execute an action.
- **State 2 (Dragging/Acting):** The system engages and modifies data.

An Always-On compass effectively lacks a "State 0"—it is permanently locked in State 1 or 2. This lack of a "neutral gear" is the theoretical root of the Midas Touch problem and induces severe psychological anxiety and a perceived loss of control (Istance et al. 2008). A "Clutch" mechanism (such as holding a thumb to the screen) restores State 0. This allows the user to explicitly disengage the system, reset their physical body posture without penalty, and re-engage, offering a definitive and biomechanically safe pause state.

2.4.3 Gross Motor Gestures and SATO

Finally, navigating hierarchical menus requires an error-recovery mechanic (a way to navigate "back" up the tree). Utilizing a physical "shake" of the device introduces a gross-motor gesture into the interface. HCI literature dictates that the physical effort required to execute an error-recovery gesture directly alters the user's adherence to the Speed-Accuracy Trade-Off (SATO) (Bi et al. 2013).

In traditional interfaces, the cost of pressing a "back" button is trivial, encouraging fast, error-prone navigation. However, because executing a gross-motor shake requires significantly more metabolic energy and introduces potential social awkwardness in public spaces, the penalty for a mistake is perceived as disproportionately high. Consequently, users will theoretically abandon rapid navigation speeds in favor of extreme, cautious accuracy, fundamentally altering the pacing of the spatial interaction (Bi et al. 2013).

3 | Design & System Development

This chapter outlines the conceptual design, architectural mapping, and technical implementation of the *Music Compass* prototype. Rather than strictly separating abstract design from code, this section details the iterative development journey, emphasizing the Human-Computer Interaction (HCI) rationale behind the system's core mechanics, redesigns, and auditory feedback loops.

3.1 System Overview and Interaction Design

The primary objective of the *Music Compass* is to facilitate eyes-free, casual music discovery using spatial proprioception. To achieve this, the system discards traditional visual hierarchies (e.g., list-based scrolling) in favour of a continuous, 360° spatial navigational paradigm.

During the initial design phase, multidimensional mappings were considered, such as utilizing intersecting 360° planes or incorporating a vertical height dimension (pitch/roll) to navigate different layers simultaneously. However, these were ultimately rejected. Utilizing a vertical axis for navigation induces severe ergonomic neck strain during prolonged use and introduces dangerous cognitive overload by violating Brewster's limits on egocentric spatial resolution (Brewster et al. 2003). Consequently, the design was deliberately constrained to a 2D circular array mapped to the user's horizontal yaw.

The user is positioned at the centre of a virtual auditory environment. The 360° space is divided into eight equal 45° sectors, each representing a specific node in the music library hierarchy. By limiting each layer to eight nodes, the system ensures distinct target acquisition while maintaining a manageable cognitive load. In this implementation, the 360-degree space is oriented horizontally on a plane around the user. By physically rotating their body (or the device), the user explores the soundscape. As the user's physical orientation aligns with a sector, the audio associated with that node smoothly cross-fades into the foreground, utilizing spatial panning to inform what node within the hierarchy has been navigated to.

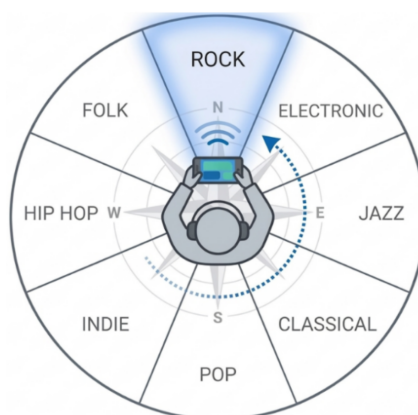


Figure 3.1: Conceptual visualization of the *Music Compass* paradigm, mapping the hierarchical auditory soundscape to a 360° physical orientation.

3.1.1 Hierarchical Navigation and the “Vertical Slice” Strategy

The conceptual music library is structured into three distinct spatial layers: **Genres** → **Artists** → **Tracks**. Navigating this three-tier hierarchy requires intuitive entry and exit interactions that do not rely on visual buttons:

- **Drilling Down (Selection):** Once a user locates a desired audio node (e.g., the “Rock” genre sector), they execute a selection command to lock in their choice. The system dynamically loads the children of that node and remaps them to the 360° compass.
- **Backtracking (The “Shake” Gesture):** To navigate backwards up the hierarchy, a gross-motor error-recovery gesture was implemented, requiring the user to physically “shake” the device. A high-pass filter isolates sharp, sudden acceleration spikes from smooth navigational rotations to trigger the backtrack state.

Implementing a complete 3-layer database across eight sectors would require actively managing 512 distinct audio files, which would overwhelm the memory limits of the prototype. Therefore, a “Vertical Slice” architectural strategy was employed. The full three-layer depth was only built out for specific evaluation targets (e.g., routing to Michael Jackson under “Pop,” or Daft Punk under “Electronic”). Non-target pathways were configured as structural “Dead Ends” that emitted a distinct error tone. This constrained scope allowed for rigorous A/B performance testing without requiring a massive, resource-heavy back-end database. **(A complete visual gallery of the prototype’s interface, including the initialization screens and all three hierarchical layers, can be found in Appendix B).**

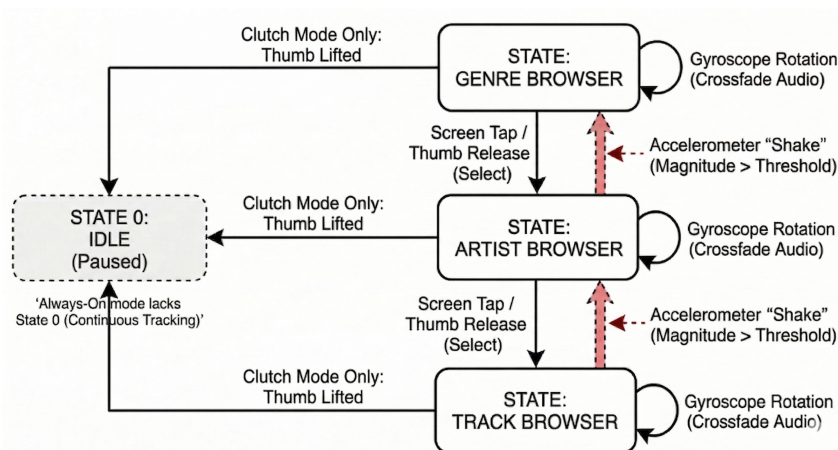


Figure 3.2: A conceptual state machine detailing the navigational hierarchy of the Music Compass. Users rotate to explore nodes, select to drill down, and perform a gross-motor shake to ascend the hierarchy.

3.1.2 Visual Interface and Experimental Scaffolding

While the *Music Compass* is fundamentally an eyes-free auditory interface, a minimal graphical user interface (GUI) was developed strictly to support the experimental evaluation phase. The screen dynamically displays a text-based “soundboard” indicating the current Genre, Artist, or Track. This visual scaffolding was crucial during the initial familiarization phase, allowing participants to visually verify the auditory cues and build their mental map. This was specifically implemented to help users acclimatize to the audio loops if they were unfamiliar with the specific artists or tracks featured in the prototype.

However, to ensure the formal evaluation tested true proprioceptive and auditory navigation rather than visual reading, a dynamic “Blur Effect” was engineered into the interface. Once a

formal evaluation task commenced, the on-screen text was heavily blurred, rendering it illegible. This successfully prevented participants from utilizing the screen as a visual crutch, forcing them to rely entirely on spatial audio and physical body orientation to locate their targets.

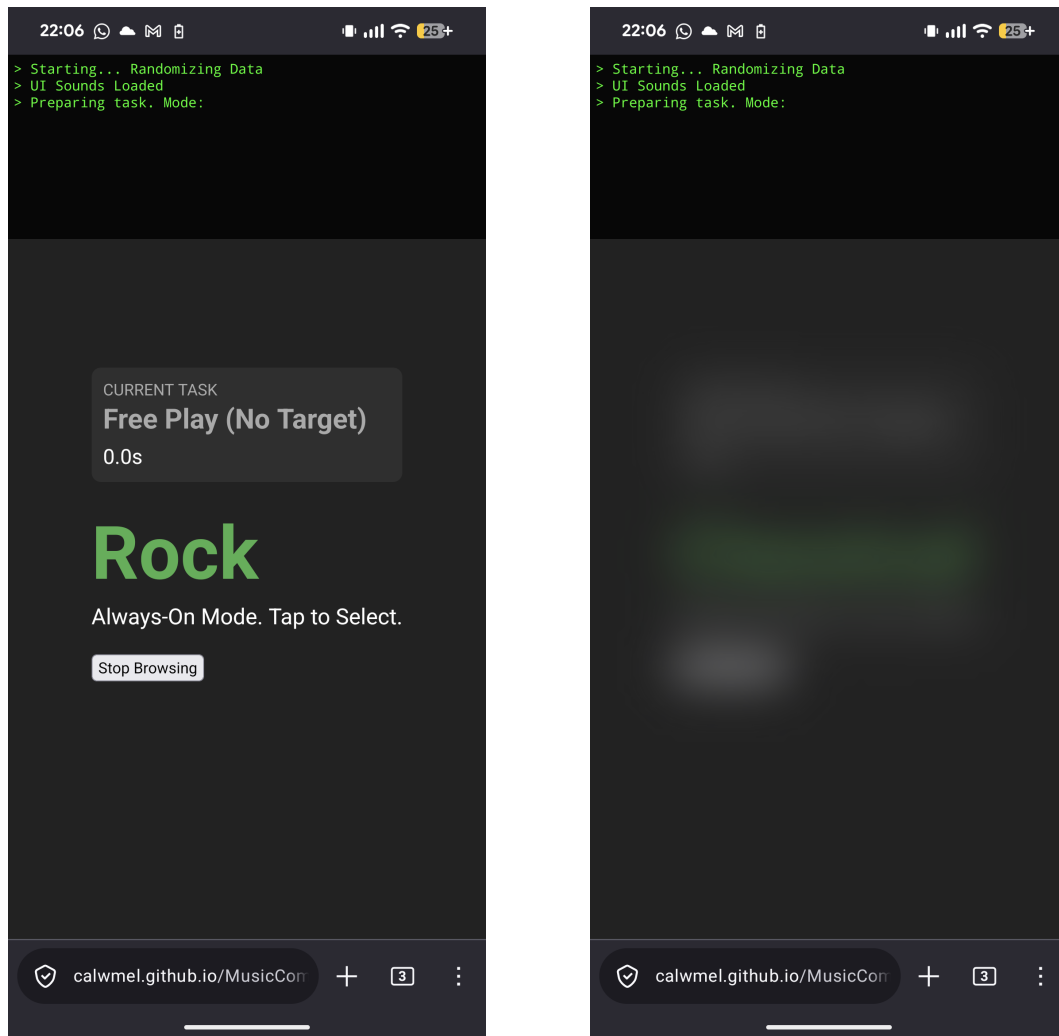


Figure 3.3: The mobile prototype interface. Left: The clear visual scaffolding utilized during the familiarization phase. Right: The dynamic blur effect applied during formal tasks to force eyes-free auditory navigation.

3.2 The Two Navigational Modalities

To directly facilitate the empirical evaluation of **RQ1 (Efficiency)** and **RQ2 (Agency)**, the prototype was engineered with two distinct navigational modalities. These modes altered how the device’s sensor data mapped to the user’s intent. To streamline the user study, these modes were not hardcoded; instead, a UI configuration toggle and an automated evaluation harness (featuring automated task timers) were integrated into the start screen, effectively turning the application into a formal research tool.

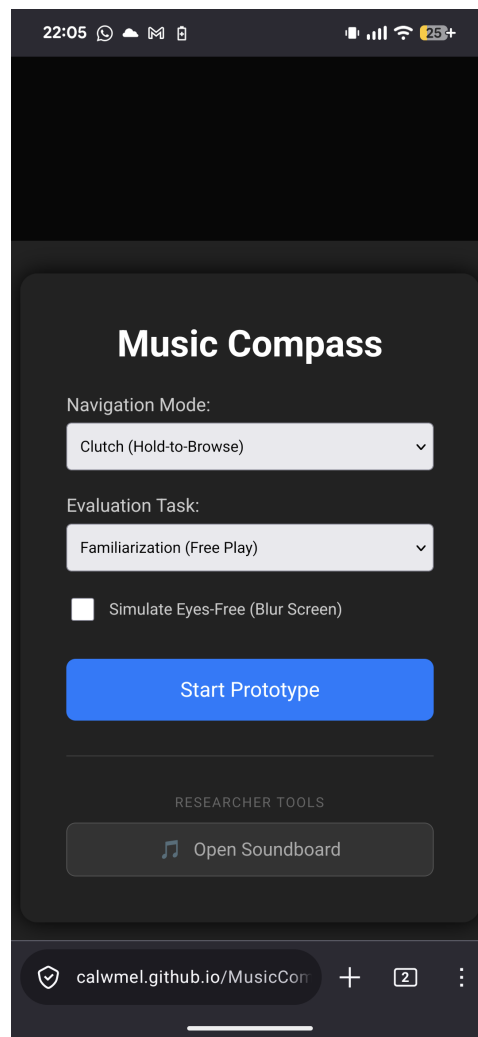


Figure 3.4: The initialization screen featuring the experimental modality toggle and evaluation harness, allowing seamless switching during the user study.

3.2.1 Condition A: The “Clutch” Mode (Hold-to-Browse)

Inspired by physical mechanical clutches, this mode requires the user to actively hold their thumb on the device screen to engage the gyroscope tracking. While the screen is depressed, physical rotation scans through the sectors defined on that particular level of the "compass". Releasing the thumb disengages the tracking and instantly selects the currently active node, subsequently pausing the exploration. This mode was designed to test the ergonomic value of an explicit pause state—meaning a physical state where the user feels completely confident the system is ignoring their movements, preventing accidental inputs.

3.2.2 Condition B: The “Always-On” Mode (Continuous Tracking)

In this modality, the device’s gyroscope is continuously polled. The user does not need to maintain physical contact with the screen to explore the 360° space; their orientation is permanently mapped to the auditory environment. To select a node, the user performs a discrete tap on the screen. This mode was designed to minimize localized grip fatigue and evaluate the theoretical efficiency

of uninterrupted spatial tracking. Crucially, an experimental visual “Stop Browsing” button was rendered on the screen during this mode. This was implemented as a fail-safe mechanism to allow users to manually halt the continuous tracking if they experienced the “Midas Touch” effect.

3.3 Auditory Design Rationale & Iterative Redesigns

Because the interface relies entirely on audio for feedback, iterative acoustic design and rigorous bug fixing were critical. Throughout the development phase, a design journal of “Audio Do’s and Don’ts” was developed by trial and error, allowing the formalization of hard User Experience rules.

- **DON’T use hard audio cuts.** Early prototypes abruptly stopped the audio of one sector and started the next when the user crossed a 45° boundary. This caused a jarring, unpleasant acoustic experience that broke user immersion.
- **DO use spatial cross-fading.** The implementation was updated to utilize continuous spatial panning. As the user approaches a sector boundary, the current audio fades smoothly while the adjacent sector’s audio fades in, utilizing the stereo field to intuitively guide the user toward the center of the next node.
- **DON’T rely on silence.** In a non-visual interface, silence is synonymous with system failure. The interface must continuously output ambient or navigational audio to assure the user the application is still tracking their movement.
- **DO utilize raw audio loops for the prototype.** To simulate a realistic music browsing experience, 10-second raw .mp3 audio loops of popular tracks were utilized. (Note: As evaluated in Chapter 5, the acoustic masking inherent in raw audio later prompted the recommendation for synthesized “Audio Earcons”).

During this iterative process, several unanticipated technical constraints regarding the Web Audio API required significant redesigns:

- **The “Swishy” 3D Audio Bug:** Early prototypes utilized the Web Audio API’s HRTF (Head-Related Transfer Function) panning model to simulate true 3D space. However, this applied complex elevation and depth filters that caused the audio to sound heavily distorted or “swishy” when positioned behind the user’s head. The implementation was downgraded to an equalpower panning model. While mathematically simpler, this solved the distortion and provided much cleaner horizontal stereo separation.
- **The “Missing Audio” Re-Decode Bug:** Navigating rapidly between layers initially caused the browser to crash and throw missing audio errors, as the system was attempting to repeatedly fetch and decode the same raw .mp3 files. This was resolved by implementing a global `bufferCache` to store decoded audio arrays in the device’s RAM, drastically reducing latency during layer transitions.
- **Audio Bleed and Pause Offsets:** In early iterations of the Clutch mode, pausing the application and resuming caused tracks to restart from the beginning. A `pauseOffset` algorithmic lock and an `isBrowsing` state flag were engineered to freeze the compass during screen taps, ensuring the audio resumed seamlessly from the exact millisecond it was paused.

3.4 Technical Implementation

To ensure cross-platform compatibility and rapid iterative deployment, the *Music Compass* was developed as a Progressive Web Application (PWA) using standard web technologies (HTML, CSS, native JavaScript).

3.4.1 Dynamic Calibration and Stereo Panning

The most significant technical hurdle involved translating raw hardware physics into a seamless acoustic experience. A smartphone's gyroscope tracks relative rotation, meaning it does not inherently know where physical "North" is. To solve this, a dynamic calibration offset was engineered. Whenever the user initiates the application, their current physical heading is mathematically captured and set to 0° . This ensures the primary "Rock" sector is always anchored directly in front of the user, regardless of their starting real-world orientation.

Furthermore, translating this 360° heading into a 3D soundscape required precise mathematical mapping. Utilizing the Web Audio API's `StereoPannerNode`, a continuous sine-wave calculation was implemented. As the user's angle shifts, the `Math.sin()` function dynamically adjusts the audio pan value between -1 (hard left) and 1 (hard right), creating the smooth, immersive cross-fade effect previously described.

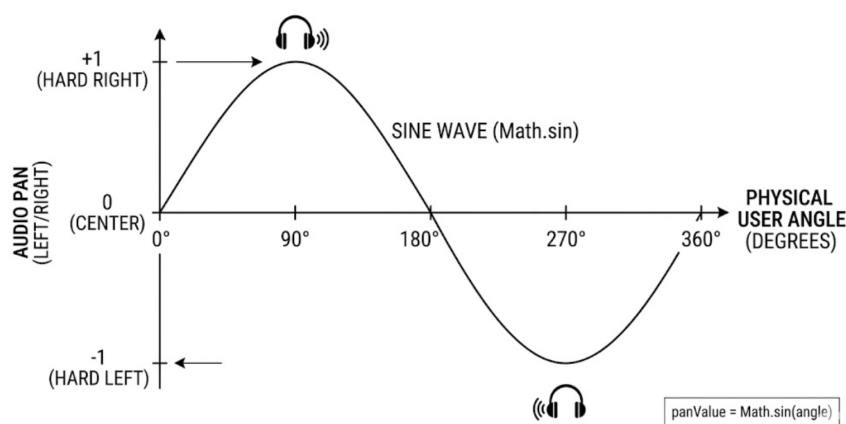


Figure 3.5: A visualization of the `Math.sin()` function mapping the user's 360° physical orientation to the stereo audio panning spectrum.

3.4.2 Sensor Integration and Gesture Detection

The application relies on Sensor Fusion, combining two distinct HTML5 APIs. The DeviceOrientation API tracks the alpha value (yaw/heading) for core compass navigation, while the DeviceMotion API tracks raw physical acceleration for the "Shake" backtrack gesture.

During development, modern browser security policies (particularly on iOS/WebKit) presented a strict constraint, as they block access to motion sensors to prevent background fingerprinting. To resolve this, a mandatory user-interaction initialization state was implemented. The application requires a definitive physical screen tap to invoke the `DeviceOrientationEvent.requestPermission()` promise, thus successfully unlocking the security sandbox before initializing the audio context.

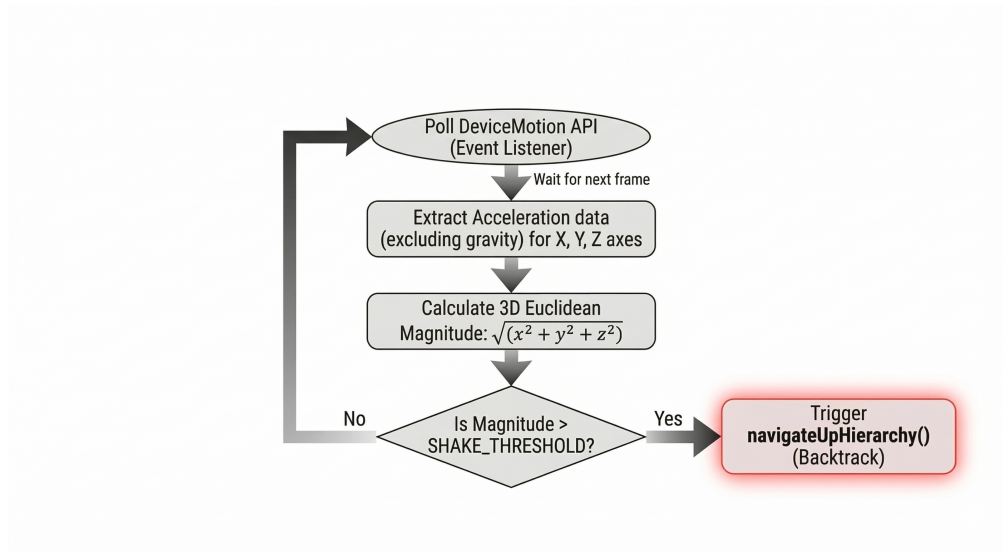


Figure 3.6: Flowchart illustrating the physics-based algorithm utilized to detect the gross-motor “Shake” gesture.

As illustrated in Figure 3.6, the gesture detection algorithm continuously polls the device’s accelerometer. By calculating the 3D Euclidean magnitude of the physical force (excluding baseline gravity), the system checks for sudden kinetic spikes. If the resulting magnitude exceeds a strictly calibrated `SHAKE_THRESHOLD`, the backtrack function is triggered.

This lightweight, browser-based architecture successfully provided the low-latency sensor polling and audio processing required to conduct the $N = 12$ formal evaluation detailed in the subsequent chapters.

4 | Evaluation Methodology & Results

To assess the objective usability and efficiency of the *Music Compass* prototype, task completion times and error rates were recorded alongside observational metrics. This chapter objectively outlines the study design and presents the quantitative and qualitative data gathered.

4.1 Evaluation Methodology

4.1.1 Objectives

The primary objective of the evaluation was to compare the efficiency and ergonomics of two distinct spatial navigation paradigms: a hold-to-browse “Clutch” mode and an “Always-On” continuous tracking mode. Furthermore, the study aimed to measure the extent to which users could build a spatial mental map of the auditory interface over sequential trials.

4.1.2 Participants

Twelve participants ($N = 12$) were recruited for the evaluation. The cohort included a mix of lay-users and Computer Science graduates, ensuring a diverse range of technical literacy and mental models regarding system architecture.

4.1.3 Experimental Design

The study utilized a **Within-Subjects Repeated Measures** design, meaning every participant tested both navigation modes. To prevent “learning bias”—where users naturally perform faster on their second attempt regardless of the mode—the cohort was perfectly counterbalanced. Six participants began with the Clutch mode (Condition A → Condition B), and the remaining six began with the Always-On mode (Condition B → Condition A).

4.1.4 Tasks & Apparatus

Evaluations were conducted with the user standing, holding a smartphone equipped with a gyroscope, and wearing headphones. Prior to the evaluation, participants were provided with an information sheet detailing the study’s purpose, a written guide to the gestural controls, and a consent form (see Appendix A). Participants then began with a two-minute “Familiarization” stage to acclimatize to the physical gestures. Following familiarization, participants executed six search tasks per mode (three Artist searches and three specific Track searches), attempting to locate targets as quickly and accurately as possible.

4.1.5 Pre-Task Scaffolding and The Soundboard

A critical anticipated confound was the varying degrees of participant familiarity with the target music. If a user did not know what Daft Punk sounded like, target acquisition would be impossible regardless of the interface’s usability. To control for this “music trivia” variable, a pre-task scaffolding protocol was engineered.

Before each trial commenced, participants were presented with a “Task Preview” screen. The researcher verbally assessed the participant’s confidence in recognizing the target genre, artist, and specific track. If the participant expressed uncertainty, the researcher utilized a custom-built “Soundboard” interface to play 5-second reference snippets of the target audio nodes. This ensured every participant had a baseline auditory mental model of their target before the spatial tracking and automated timers were engaged.

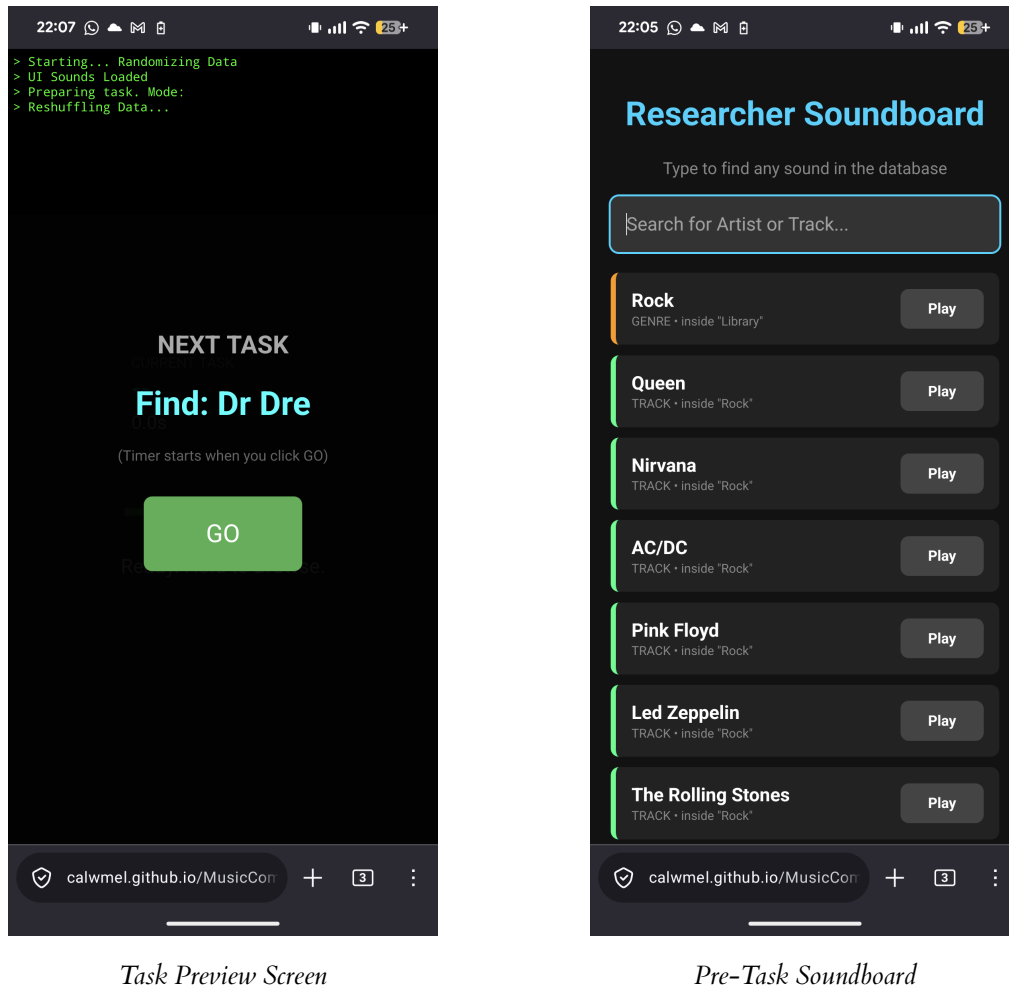


Figure 4.1: The experimental scaffolding tools used to prime participants’ auditory memory prior to formal task execution.

4.1.6 The Visual “Stop” Button

During the Always-On condition, an experimental visual “Stop Browsing” button was rendered directly on the screen. This was implemented as a fail-safe mechanism to allow users to manually halt the continuous tracking if they experienced severe disorientation.

4.1.7 Metrics Collected

To provide a holistic analysis of the interaction paradigms, both objective performance data and subjective user experience metrics were systematically recorded.

Objective Quantitative Metrics:

- **Task Completion Time:** Measured in seconds from the initiation of a search task to the successful acquisition of the target node. This was tracked across three repeated Artist searches and three Track searches per mode to quantify the learning curve.
- **Error Rates (Backtracks):** Errors were not measured by standard misclicks, but by the number of physical “Shake” gestures a participant had to execute to recover from an incorrect node selection.

Subjective Qualitative Metrics: Following the completion of the physical tasks, semi-structured interviews were conducted to capture the user’s psychological and ergonomic experience. The interview protocol specifically targeted:

- **Spatial Mental Mapping:** Participants were asked to blindly estimate the depth (number of layers) and breadth (items per ring) of the interface to assess how well they constructed an invisible mental model of the system architecture.
- **Ergonomics and Physical Fatigue:** Assessing the trade-off between localized thumb/grip fatigue (Clutch mode) and broader postural/arm fatigue (Always-On mode).
- **Perceived Controllability:** Evaluating which mode provided a higher sense of user agency and precision.
- **Gesture Naturalness:** Gathering feedback on the physical effort, satisfaction, and false-positive rate of the gross-motor “Shake” mechanic.
- **The Learning Curve:** Measuring the user’s perceived cognitive load, specifically assessing whether the interaction shifted from conscious, effortful movement to automatic muscle memory over the course of the trials.

4.2 Quantitative Results

Formal inferential statistics (Paired-Samples t-tests and Wilcoxon Signed-Rank tests) were applied to the data log to determine the mathematical significance of the observed trends.

4.2.1 Efficiency and Task Completion Times

The primary metric for efficiency was the average time taken to successfully locate a target item. A paired-samples t-test revealed a statistically significant difference in task completion times ($t(11) = 4.61, p < 0.001$). Participants were significantly faster using the Always-On mode ($M = 29.39s, SD = 8.96s$) compared to the Clutch mode ($M = 43.22s, SD = 15.99s$).

Notably, 11 out of the 12 participants recorded faster overall averages in the Always-On mode. The data also highlighted massive variance in extreme cases: P1 achieved a target acquisition in just 6.5 seconds during an Always-On trial, proving that spatial muscle memory allows for near-instantaneous selection. Conversely, P12 required 187.2 seconds to find a specific track in Clutch mode due to disorientation. The 95% confidence interval plot (Figure 4.2) visually supports this efficiency gap.

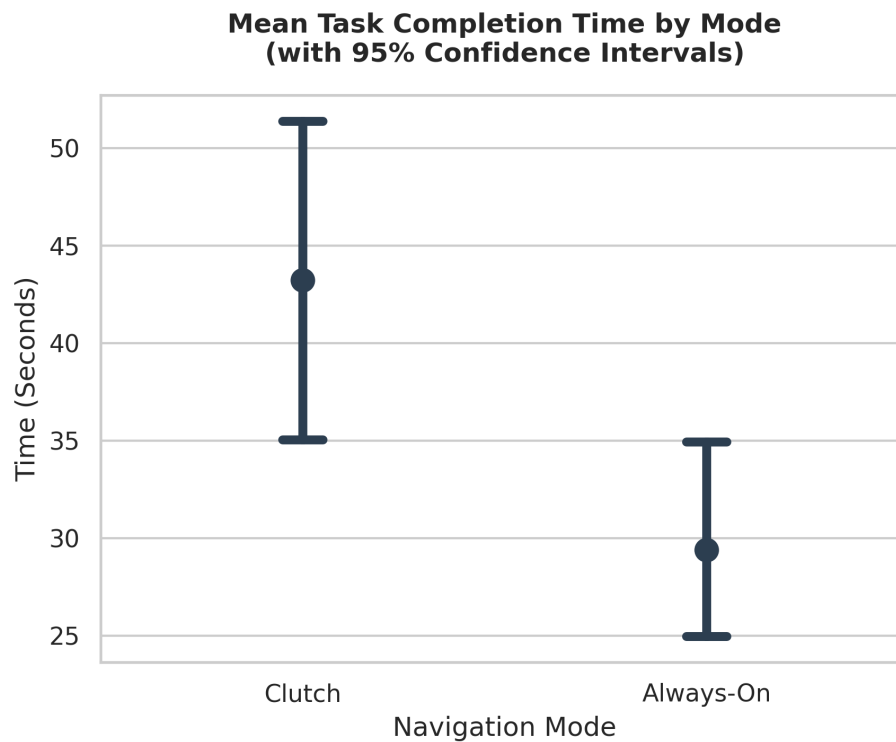


Figure 4.2: Mean task completion time by mode with 95% confidence intervals.

4.2.2 The Learning Effect and Spatial Mental Mapping

To analyse spatial motor learning, the average completion times for the identical Artist Search tasks were plotted chronologically (Trial 1 → Trial 2 → Trial 3). As shown in Figure 4.3, a downward trend is visible across both modes. In the Clutch condition, the average search time dropped from 45.1s in Trial 1 down to 19.6s by Trial 3.

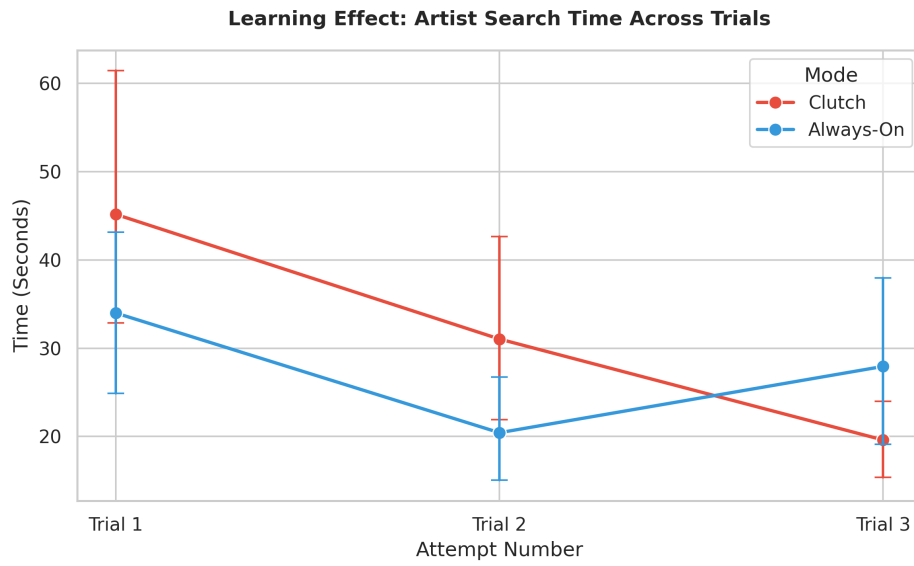


Figure 4.3: Learning Effect: Artist Search Time across consecutive trials.

4.2.3 Error Rates

Navigation errors were measured by counting the number of times a user had to execute the “Shake” gesture to backtrack. Overall error rates were low (Clutch $M = 2.75$ errors; Always-On $M = 1.75$ errors). A Wilcoxon Signed-Rank Test yielded $W = 11.0, p = 0.085$, indicating the difference between the modes is not statistically significant at the $\alpha = 0.05$ level (Figure 4.4).

During observation, participants were noted to move highly deliberately and cautiously during the trials, actively attempting to avoid performing the “Shake” gesture.

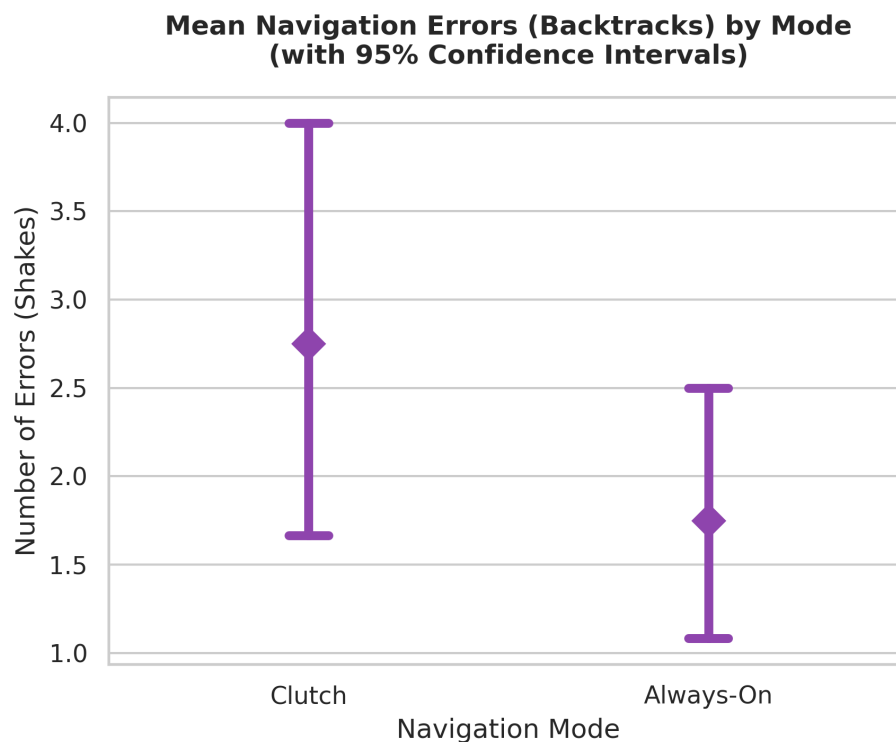


Figure 4.4: Mean navigation errors (backtracks) by mode.

4.3 Qualitative Results

Post-task interviews and observational notes provided subjective data on the ergonomics, acoustics, and usability of the prototype.

4.3.1 Physical Fatigue and Interaction Ergonomics

When reporting physical exertion, responses were polarized based on the mode. P5, P6, P9, and P11 noted that the Always-On mode induced postural fatigue due to the requirement of holding the device continuously. Conversely, P1 and P7 reported localized thumb fatigue in Clutch mode.

The physical environments and postures that participants chose to use during the evaluation varied slightly. While instructed to stand, some participants rotated their entire body, others solely rotated the device around their torso, and one participant actively utilized a swivel chair to navigate. Interestingly, observational data showed several participants spontaneously removed one ear of their headphones during the trials, keeping one ear open to the room.

4.3.2 Perceived Controllability and the Ignored Stop Button

Despite being statistically slower, the majority of participants stated that the Clutch mode felt more precise. Participants cited the affordance of lifting their thumb as a reason for this preference. P3 noted that with the Clutch mode, “if you see your mates... you can literally just take a thumb off” to stop the audio.

Conversely, during the Always-On trials, many participants operated under the false impression that the mode completely lacked a stop mechanism. Observational data confirmed that participants entirely ignored the visual “Stop Browsing” button rendered on the screen, skewing their preference heavily toward the physical safety of the Clutch mode.

4.3.3 The Efficacy of the Soundboard

Because the prototype used the raw intros of commercial tracks, participants frequently relied on the pre-task Soundboard to prime their memory. However, observational data revealed this scaffolding had mixed success. Several participants exhibited overconfidence, declining the Soundboard preview only to become severely lost during the task when they failed to recognize the slow instrumental build-ups.

Conversely, even participants who actively listened to the Soundboard previews still frequently struggled to isolate those specific tracks when placed in the immersive 360° environment, reporting that genres with similar acoustic profiles blended together. Multiple participants stated they would vastly prefer to use the system with their own customized, familiar playlists.

4.3.4 Gamification and Entertainment Value

A recurring, unprompted theme was the inherent “fun factor” of the spatial interaction. P3 and others explicitly likened the spatial search mechanics to an auditory guessing game, comparing the experience to popular music-trivia applications like *Heardle*, but enhanced by the physical, embodied nature of the compass.

4.3.5 Feedback on Gesture Mechanics

When discussing the physical “Shake” gesture used to backtrack, opinions were divided. While some found it satisfying, P6 stated, “I thought I was going to throw the phone... I had to actually jerk my wrist.” Consequently, several participants independently suggested replacing the gross-motor shake with a more discrete “flicking” motion utilizing the gyroscope.

5 | Discussion & Future Work

This chapter synthesizes the quantitative results and qualitative observations from Chapter 4 to interpret *why* specific interaction patterns occurred. By cross-referencing these findings with established Human-Computer Interaction (HCI) literature, this section critically analyses the vagaries of the systems tested, details unanticipated emergent behaviours, and utilizes these insights to propose concrete, evidence-based future work.

5.1 Synthesizing Usability & Performance

5.1.1 Gross Motor Gestures and the Speed-Accuracy Trade-Off (SATO)

The quantitative data (Figure 4.4) revealed an absence of statistical variance in error rates between the modalities, despite their differences and the complexity of the spatial task. When taken with the observational transcripts, this showed the evaluators intentionally slowed down to actively avoid errors. This is consistent with a critical adaptation of the *Speed-Accuracy Trade-off* (SATO) within mobile gesture interfaces (Bi et al. 2013).

In traditional desktop computing, the penalty for an error (e.g., pressing backspace) is trivial, leading users to optimize their behaviour for speed. However, the *Music Compass* prototype utilized a gross-motor gesture (a physical “shake”) as the error-recovery mechanism. Because participants found the physical shake penalty to be socially awkward or physically disruptive to their navigational flow (with P6 fearing they might “throw the phone”), users actively subverted their own navigation speed.

This behavioral shift is highly significant within the context of HCI research; it demonstrates that gestural mobile interfaces cannot be evaluated using traditional, desktop-oriented error models. The metabolic and social costs of gross-motor gestures fundamentally bottleneck interaction speed, forcing users to deliberately trade efficiency to ensure high accuracy (Robinson 2010).

5.1.2 Proprioception and Spatial Motor Learning

Despite this cautious baseline behaviour, the rapid trial-over-trial decrease in task completion time (Figure 4.3) provides robust evidence of spatial motor learning. As explicitly noted by P10, by the final trial, users were relying less on sequential audio feedback and instead utilizing proprioception—physically rotating their body to a known spatial angle (e.g., 90°) to instantly select a target.

This confirms the primary hypothesis: users can successfully build robust spatial mental maps of non-visual interfaces, allowing physical muscle memory to bypass the need for continuous auditory prompting.

5.2 Affordances, Agency, and Mental Models

A critical discrepancy emerged between the objective performance and subjective preference of the two navigation modes. While the Always-On mode was statistically faster ($p < 0.001$), participants

overwhelmingly felt the Clutch (hold-to-browse) mode was “safer” and more controllable.

This preference illustrates a known paradigm conflict in sensor-based HCI (Istance et al. 2008). While “always-on” tracking appears more seamless conceptually, it inherently lacks a State 0 (Out of Range) interaction phase, frequently inducing an illusion of lost control known as the “Midas Touch” problem. Users felt secure using the Clutch mode because lifting their thumb provided an instant, tactile method to pause the interaction, offering biomechanical assurance.

Crucially, the experimental visual “Stop” button included in the Always-On mode failed to provide this assurance because participants entirely ignored it. This proves that once a user’s mental model shifts to an embodied, eyes-free proprioceptive paradigm, visual screen affordances become functionally invisible (Li 2024). The Clutch mode’s subjective victory is therefore derived from the *ergonomics of agency*—its unambiguous, physically afforded “pause” state restored the user’s perceived control, a fundamental requirement for interfaces used in unpredictable environments (Istance et al. 2008).

5.3 Context of Use & Unanticipated Outcomes

Some of the most interesting insights from the evaluation emerged from unprompted user behaviours, revealing critical design considerations for the deployment of spatial audio systems in “Nomadic HCI” scenarios (Sarsenbayeva et al. 2017).

5.3.1 Auditory Masking vs. Musical Familiarity

During track-finding tasks, completion times increased significantly. Interviews revealed this was largely due to a lack of initial *auditory distinctiveness*. Because the prototype utilized raw audio files, participants frequently struggled to quickly differentiate between certain genres with overlapping acoustic profiles (e.g., Pop vs. Electronic) based solely on the first few seconds of a track intro.

Crucially, the mixed success of the pre-task “Soundboard” highlights a fundamental limitation in human cognitive processing. While the Soundboard successfully loaded the target audio into the user’s short-term phonological loop, this temporary memory was fragile and easily overwritten by the cognitive overload of the spatial panning task. Hearing a snippet just before a task is often insufficient to combat active auditory masking (McGookin and Brewster 2004).

However, participants noted that this issue would largely vanish if the system allowed for playlist customization. This highlights a powerful psychological mechanic: substituting unknown audio with highly familiar, customized music shifts the cognitive load from short-term *recall* to deeply ingrained, long-term *recognition*. This drastically accelerates spatial target acquisition because the user is no longer actively listening to decipher the track; they are simply reacting to a known, familiar stimulus (Schubhan et al. 2024).

5.3.2 Nomadic HCI and Environmental Interference

A core assumption of the *Music Compass* was that users would map audio to a continuous 360° space anchored to their physical body. However, P9 raised a critical flaw regarding environmental interference in non-holonomic transit environments (e.g., a turning bus). Because the system relies entirely on the device’s localized gyroscope, the physical turning of a vehicle inadvertently rotates the user’s entire audio space, causing severe spatial disorientation even if the user remains completely still.

5.3.3 Situational Awareness and Physical Adaptations

Despite the transit limitations, the evaluation proved the system successfully accommodates pedestrian situational awareness. Multiple participants were observed actively subverting the noise-isolating nature of the headphones by keeping one earphone off to monitor their physical surroundings (Sarsenbayeva et al. 2017). Furthermore, users spontaneously adapted the physical interaction to minimize motor fatigue (e.g., swiveling in an office chair). By transferring the rotation from the torso to the wrist or a chair, the interface accommodates seated or mobility-impaired users, bypassing the need for fine-motor touch controls.

5.4 Future Work

The unanticipated outcomes and limitations discovered during this evaluation directly inform three specific, evidence-based areas for future iteration.

- **Acoustic Distinctiveness and Personalization:** To resolve the “music trivia” confound and mitigate the cognitive delay caused by auditory masking, future iterations must employ a two-tiered approach to audio feedback. For high-level menu scanning (e.g., Genres), raw commercial intros should be replaced with synthesized “Audio Earcons”—highly stylized, brief auditory icons that enable rapid spatial target acquisition (Walker et al. 2006). Conversely, for deeper track-level navigation, the system should integrate with established streaming APIs (e.g., Spotify). Allowing users to populate the spatial sectors with their own personalized playlists shifts the cognitive demand from recall to rapid recognition memory, thereby vastly improving navigational confidence and speed (Garcia and Neris 2013).
- **Gamification of Spatial Audio:** Given the unprompted enjoyment participants derived from locating hidden tracks, the interface paradigm could easily be adapted into a gamified, location-based music trivia experience (analogous to an embodied, spatial version of *Heardle*). Utilizing spatial audio for gamification represents an entirely untapped market for interactive audio entertainment, drastically increasing user engagement and motivation (Poudratchi et al. 2024; Aliady and Poesio 2026).
- **The 180-Degree “Rolodex” Design:** To resolve the gyroscope interference in moving vehicles and accommodate crowded transit spaces, the continuous 360° compass should be transitioned to a discrete 180° “Rolodex” design. Users could utilize subtle, directional gyro-flicks (e.g., flicking the wrist left or right) to cycle through an auditory space anchored directly in front of their forward-facing field of view.



Figure 5.1: Conceptual sketch of the discrete 180-degree “Rolodex” interaction paradigm, addressing non-holonomic transit interference.

- **Spatial and AR Anchoring:** To capitalize on the strong proprioceptive memory observed during the trials, future iterations could incorporate Augmented Reality (AR) spatial anchoring. As suggested by P7, users could persistently map specific genres to physical locations in their own environment (e.g., anchoring a “Rock” playlist to a specific wall poster in their bedroom). This would transition the interface from a relative egocentric map to an absolute allocentric map, leveraging physical landmarks for persistent audio navigation.

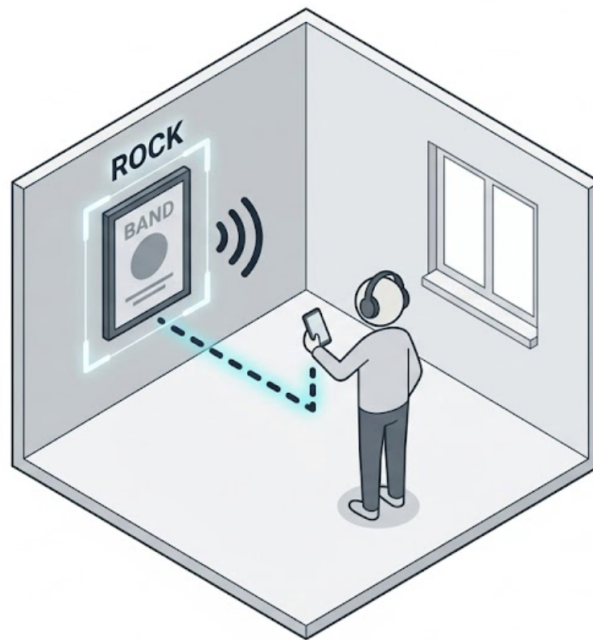


Figure 5.2: Conceptual sketch of Augmented Reality (AR) anchoring, mapping digital audio nodes to physical real-world landmarks.

6 | Conclusion

The overarching goal of this dissertation was to address the visual bottleneck in nomadic mobile computing by designing, implementing, and formally evaluating a non-visual prototype of a hierarchical spatial audio menu. Through the integration of iterative software design and rigorous formal evaluation, this goal was successfully achieved.

To test the viability of spatial proprioception, the *Music Compass* was successfully developed. Utilizing modern smartphone Inertial Measurement Units (IMUs) and the HTML5 Web Audio API, a Progressive Web Application (PWA) was engineered to map a music library to the physical space surrounding the user. Throughout the iterative design process, several complex architectural challenges were overcome. Multidimensional interaction models (such as intersecting 360° planes or vertical height axes) were explicitly considered but ultimately rejected to prevent ergonomic neck strain and cognitive overload. Furthermore, to bypass browser memory constraints during audio decoding, a “Vertical Slice” data architecture was implemented. This resulted in a refined, highly responsive prototype that successfully anchored a three-tier music library to a continuous 2D horizontal plane.

This prototype was utilized to conduct a counterbalanced, $N = 12$ within-subjects evaluation. The study was designed to formally compare two distinct interaction modalities: an “Always-On” continuous tracking mode, and an explicit “Clutch” (hold-to-browse) mode. By comparing these paradigms, the study definitively answered the three core research questions established at the outset of this project:

- **RQ1 (Efficiency and Target Acquisition):** The quantitative evaluation proved that continuous tracking (Always-On mode) results in statistically faster target acquisition within a spatial audio menu ($p < 0.001$).
- **RQ2 (Usability and Agency):** Despite being mathematically slower, users overwhelmingly preferred the explicit “Clutch” mode. The findings demonstrated a profound requirement for the *Ergonomics of Agency*—the presence of an explicit, tactile pause state provides vital biomechanical assurance, neutralizing the anxiety of the “Midas Touch” problem and vastly increasing the user’s perceived sense of control.
- **RQ3 (Error Recovery and SATO):** The inclusion of a gross-motor “Shake” gesture for error recovery successfully altered the user’s Speed-Accuracy Trade-off (SATO). To avoid the physical and social penalty of executing the gross-motor gesture, users proactively slowed their navigational pacing to ensure high accuracy, resulting in statistically non-significant variance in error rates across both modalities.

Beyond the quantitative metrics, conducting this evaluation captured rich qualitative data and surfaced several unanticipated emergent behaviours. For instance, participants were observed actively subverting the noise-isolating headphones by keeping one ear open, empirically validating the necessity of eyes-free interfaces for maintaining pedestrian situational awareness. Furthermore, users spontaneously adapted the spatial interface to minimize postural fatigue, such as navigating via a swivel office chair. While this proved the navigational framework is highly flexible and accessible for seated users, it highlighted a limitation in the evaluation’s environmental constraints.

Reflecting on the project lifecycle, if this evaluation were to be repeated, the testing environments would be more strictly defined and controlled (e.g., enforcing uniform postural constraints

across the entire cohort). Because users deliberately slowed down to avoid the “Shake” penalty, baseline error rates were difficult to differentiate between the two modalities. Implementing a time-constrained “stress test” would force rapid navigation and expose the true error thresholds of the paradigms. Furthermore, the evaluation utilized raw audio clips to mimic a real music player, which suffered from auditory masking and lacked immediate cognitive distinctiveness. Allocating early development time to synthesize stylized “Audio Earcons” would have eliminated this variable entirely. Finally, conducting contextual “in-the-wild” pilot testing earlier in the development cycle (e.g., on a moving bus) would have surfaced non-holonomic transit limitations sooner, validating the proposed discrete “Rolodex” design prior to the formal evaluation.

Despite these reflective limitations, the project successfully navigated complex web-audio engineering constraints to directly address a critical accessibility gap in mobile computing. The *Music Compass* successfully proved that human proprioception and spatial motor learning offer a highly intuitive, high-bandwidth mechanism for eyes-free navigation. Ultimately, this project achieved its core research goals, delivering both a robust technical proof-of-concept and a critical, evidence-based exploration into the future of spatial computing.

A | Evaluation Materials

This appendix contains the supplementary documents utilized during the formal $N = 12$ user evaluation, ensuring adherence to the University's ethical guidelines.

Participant Information Sheet

Project Title: Evaluation of "Music Compass": A Non-Visual Spatial Audio Interface

Researcher: Calum Woods

Supervisor: Euan Freeman

What is the purpose of this study?

You are invited to participate in a study evaluating a new way to browse music on a mobile phone without looking at the screen. This system, "Music Compass," uses spatial audio (3D sound) and physical gestures (rotation and tilting) to help you navigate a music library. The goal is to compare two different interaction modes to see which is more intuitive.

What will I be asked to do?

1. **Familiarization:** You will be given a few minutes to learn the gestures and explore the system. (You will be encouraged to look at the screen during this phase to verify what you are hearing).
2. **Tasks:** You will be asked to locate specific songs (e.g., "*Find Billie Jean*") and answer questions about the library structure (e.g., "*How many artists are in the Pop genre?*"). You will perform these tasks using two different interaction modes.
3. **Interview:** At the end, we will have a short discussion about your experience.

What about the music?

Please Note: This is a research prototype, not a complete music app. To save time, only specific "vertical slices" of the library are fully populated with tracks (specifically the **Pop** and **Electronic** genres (Micheal Jackson and Daft Punk)).

Voluntary Participation

Your participation is entirely voluntary. You may withdraw from the study at any time without giving a reason. Your data will be anonymized and used solely for this dissertation project.

Prototype Guide

How to Control "Music Compass"

This interface relies on **Hearing** and **Feeling**, not seeing. You will try two different modes:

1. The Modes (How to Browse)

Mode A: "Clutch" Mode

HOLD YOUR THUMB on the screen to engage the engine.

While holding, physically **ROTATE** the phone around you to hear different items.

If you **release** your thumb, the sound stops.

Mode B: "Always-On" Mode

The engine is **ALWAYS ACTIVE**.

Simply **ROTATE** the phone around you to hear different items.

If you want to stop/pause the browsing, **TAP** the screen once.

2. The Layers (The Structure)

The library is arranged in a hierarchy.

You always start at the top. Going deeper requires a tap on the screen.

- **Layer 1: Genres** (Select one to see Artists)
- **Layer 2: Artists** (Select one to see Tracks)
- **Layer 3: Tracks** (Select one to Play/Lock)

3. The Selection (How to Select & Listen)

- **TAP** the screen (quick touch) to **LOCK** your selection, going deeper in the hierarchy.
- **Crucial for the Track Layer:** When you find the song you want, **TAP** to lock it. This allows you to put the phone down or move around and listen to the full song without the audio switching to the next sector.
- *Note:* Do not tap if you are trying to browse. Tapping locks you in place.

4. Going Back

- **SHAKE** the phone firmly to go back up a level.
 - *Example:* If you are looking at **Tracks**, shake to go back to **Artists**.

Consent form



Music Compass: Designing and Evaluating a Spatial-Audio Interface for Non-Visual Music Browsing on Mobile Devices

Calum Woods
School of Computing Science
University of Glasgow
2602747w

Supervisor: Euan Freeman
euan.freeman@glasgow.ac.uk

Please read the following statements carefully. If you agree with each statement, please tick (✓) in the space provided. If you have any questions about what you are being asked to provide consent to, please ask.

- | | |
|--|-------|
| I am voluntarily agreeing to participate in this research study. | _____ |
| I have been given an information sheet describing this study. | _____ |
| I understand that I may withdraw from this study at any time, without giving a reason. | _____ |
| I understand that data gathered during this study is anonymous. | _____ |
| I understand that anonymous data from this study may be shared in open-access data repositories. | _____ |
| I understand that if I withdraw, any gathered data will be destroyed. | _____ |
| I understand that my voice will be recorded and processed using transcription software. | _____ |

Your signature:

Date: _____ / _____ / _____

**School of Computing Science
University of Glasgow**

Ethics checklist form for 3rd/4th/5th year, and taught MSc projects

This form is only applicable for projects that use other people ('participants') for the collection of information, typically in getting comments about a system or a system design, getting information about how a system could be used, or evaluating a working system.

If no other people have been involved in the collection of information, then you do not need to complete this form.

If your evaluation does not comply with any one or more of the points below, please contact the Chair of the School of Computing Science Ethics Committee (matthew.chalmers@glasgow.ac.uk) for advice.

If your evaluation does comply with all the points below, please sign this form and submit it with your project.

-
1. Participants were not exposed to any risks greater than those encountered in their normal working life.

Investigators have a responsibility to protect participants from physical and mental harm during the investigation. The risk of harm must be no greater than in ordinary life. Areas of potential risk that require ethical approval include, but are not limited to, investigations that occur outside usual laboratory areas, or that require participant mobility (e.g. walking, running, use of public transport), unusual or repetitive activity or movement, that use sensory deprivation (e.g. ear plugs or blindfolds), bright or flashing lights, loud or disorienting noises, smell, taste, vibration, or force feedback

2. The experimental materials were paper-based, or comprised software running on standard hardware. *Participants should not be exposed to any risks associated with the use of non-standard equipment: anything other than pen-and-paper, standard PCs, laptops, iPads, mobile phones and common hand-held devices is considered non-standard.*

3. All participants explicitly stated that they agreed to take part, and that their data could be used in the project.

If the results of the evaluation are likely to be used beyond the term of the project (for example, the software is to be deployed, or the data is to be published), then signed consent is necessary. A separate consent form should be signed by each participant.

Otherwise, verbal consent is sufficient, and should be explicitly requested in the introductory script.

4. No incentives were offered to the participants.

The payment of participants must not be used to induce them to risk harm beyond that which they risk without payment in their normal lifestyle.

5. No information about the evaluation or materials was intentionally withheld from the participants.
Withholding information or misleading participants is unacceptable if participants are likely to object or show unease when debriefed.
6. No participant was under the age of 16.
Parental consent is required for participants under the age of 16.
7. No participant has an impairment that may limit their understanding or communication.
Additional consent is required for participants with impairments.
8. Neither I nor my supervisor is in a position of authority or influence over any of the participants.
A position of authority or influence over any participant must not be allowed to pressurise participants to take part in, or remain in, any experiment.
9. All participants were informed that they could withdraw at any time.
All participants have the right to withdraw at any time during the investigation. They should be told this in the introductory script.
10. All participants have been informed of my contact details.
All participants must be able to contact the investigator after the investigation. They should be given the details of both student and module co-ordinator or supervisor as part of the debriefing.
11. The evaluation was discussed with all the participants at the end of the session, and all participants had the opportunity to ask questions.
The student must provide the participants with sufficient information in the debriefing to enable them to understand the nature of the investigation. In cases where remote participants may withdraw from the experiment early and it is not possible to debrief them, the fact that doing so will result in their not being debriefed should be mentioned in the introductory text.
12. All the data collected from the participants is stored in an anonymous form.
All participant data (hard-copy and soft-copy) should be stored securely, and in anonymous form.

Project title Music Compass

Student's Name Calum Woods

Student Number 2702747w

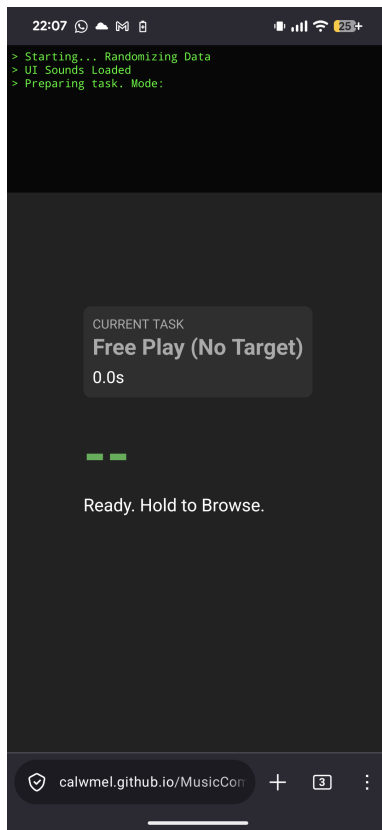
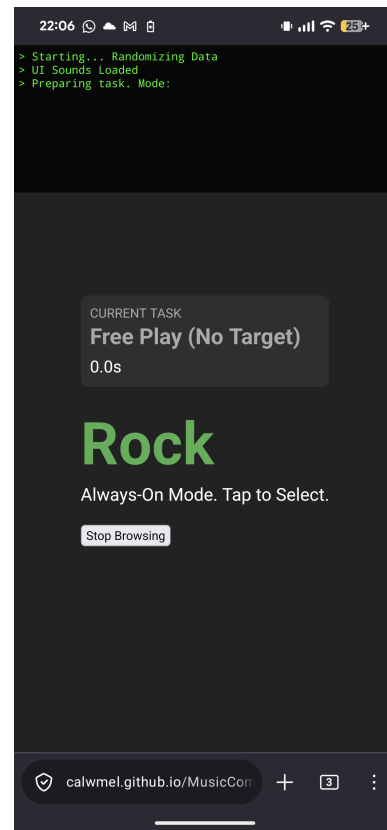
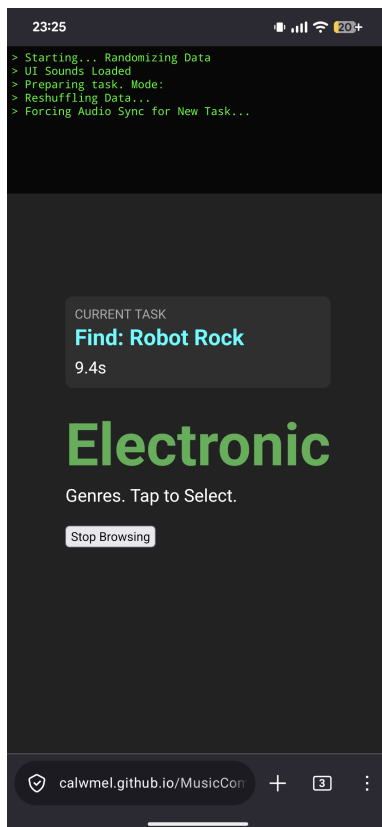
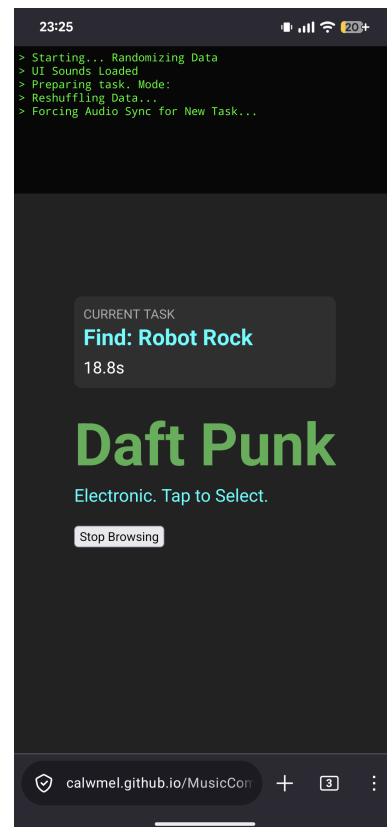
Student's Signature Calum W

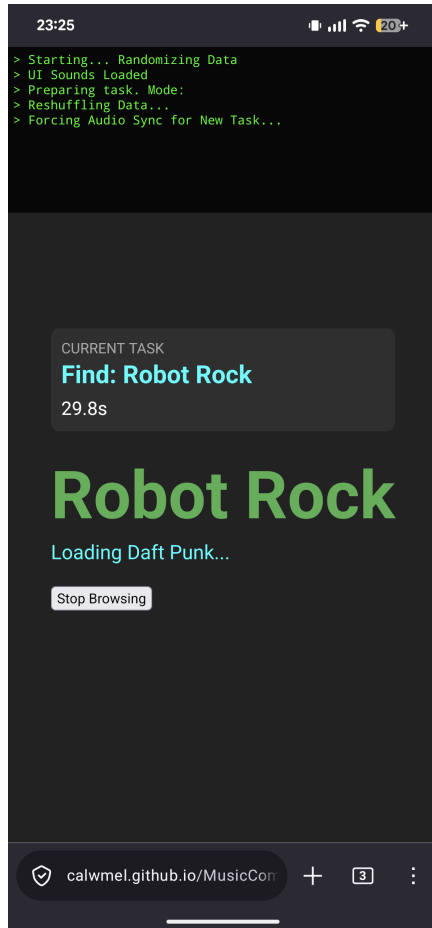
Supervisor's Signature E. Freeman

Date 26/3/20

B | Prototype Interface Gallery

This appendix provides a comprehensive visual overview of the graphical user interface, including the experimental mode toggles and the visual scaffolding for each hierarchical layer.

*Clutch Mode Initialization**Always-On Mode Initialization**Layer 1: Genre Selection**Layer 2: Artist Selection*



Layer 3: Track Selection

Figure B.1: Progression through the hierarchical layers and mode selection screens of the Music Compass prototype.

C | System Deployment & Readme

C.1 Source Code Overview

The complete source code for the *Music Compass* prototype, along with the `Master Data Log.csv` containing the raw evaluation metrics, is provided in the supplementary `.zip` file submitted alongside this dissertation.

C.2 How to Run the Prototype

The most reliable and immediate method to test the *Music Compass* prototype is via its live web deployment.

Live Deployment

1. On a mobile device, navigate to: <https://calwmel.github.io/MusicCompass/>
2. *Note: The system has been evaluated primarily using Mozilla Firefox on Android (Google Pixel 7 Pro).*
3. Tap the screen to initialize the Audio Context and grant the browser permission to access the device's motion and orientation sensors when prompted.

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